

The Rough-Heston Call as One Absolutely-Convergent erfc-Hermite Series

Grounded in provable mathematics, not in the present (broken) source — with
a verified stopping rule

arb4j

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0. Standing of this document

Everything below is grounded in **what is mathematically provable**, derived from the *definition* of the option price and standard complex analysis. The present artifacts — the Java in `arb.functions.*`/`arb.expressions.*` and the LaTeX in `docs/frmp.tex` — are treated as **unverified and currently wrong**: they are not used as authorities, and no step here is justified by “the source does it this way.” Where a fact also happens to appear in `frmp.tex`, it is **re-proved from scratch here** and, for the three load-bearing identities, **numerically confirmed to machine precision** (the certificates are quoted in §5, §7). The test of correctness is mathematical — the exact Black-Scholes limit, a proven geometric tail bound, and an independent numerical cross-check against the defining contour integral — never agreement with the source.

The deliverable the user asked for: **one** infinite series (single index n , not a multi-index lattice) for the call price, with the **erfc/Hermite** kernel, **no assumed pole structure**, an **explicit error bound that is the stopping rule**, and a proof of convergence. It exists, it is derived and proved below, and it was validated end-to-end (err 1.1×10^{-14} at 40 terms; §7.3).

1. The price as a mathematical object, and what is actually known

Let $X_T = \log(S_T/S_0) - rT$ be the discounted log-return and $\phi_T(v) = \mathbb{E}[e^{ivX_T}] = e^{\Phi(v,T)}$ its characteristic function. Fix moneyiness $\tilde{k} = \log(K/S_0) - rT$, so $K = S_0 e^{rT + \tilde{k}}$. The discounted European call is, *by definition*,

$$C = e^{-rT} \mathbb{E}[(S_T - K)^+] = e^{-rT} \mathbb{E}[g_C(X_T)], \quad g_C(x) = S_0 e^{rT} (e^x - e^{\tilde{k}})^+. \quad (1.1)$$

Two distinct claims must never be conflated; the present source conflates them and that is part of why it is “not known to produce the correct answer.”

- **(U) Unconditional.** *For any rational surrogate exponent $\Phi_M(v) = (\text{polynomial}) + (\text{proper rational})$ with strictly negative quadratic symbol, the price under the surrogate $\phi_M = e^{\Phi_M}$ is exactly the single series of §6, with the verified error bound of §7. This is a theorem of complex analysis; it depends on nothing about rough Heston, nothing about Padé, and no pole locations.*

- **(C) Conditional.** That surrogate ϕ_M approximates the true rough Heston ϕ_T . This is the El Euch–Rosenbaum analyticity of Φ plus Stahl convergence of the diagonal Padé of the Müntz series $\tilde{\Phi}(z; v) = \sum_k d_k(v) z^k$ ($z = T^\mu$, $\mu = H + \frac{1}{2}$). It is conditional (capacity convergence, generic non-degeneracy) and is **not** this report's subject.

The user's instruction — single sum, no assumed poles — is precisely the demand to deliver **(U)** cleanly and stop smuggling **(C)**'s asymptotic pole-location claims into the price formula. Everything below proves **(U)**.

The only structural input used about the surrogate is:

(H) A contour exists. There is $c \in (1, p^*)$ with $\phi_M(-iw)$ finite on $\Re w = c$, where p^* = critical moment. Equivalently the proper part of Φ_M has no pole on or left of the line $\Re w = c$.

(H) is *necessary*: without it the price integral (2.1) diverges (moment explosion). It is the genuine, checkable integrability condition. It is **not** the assertion that the surrogate has $2M$ simple poles all in the lower half-plane — that assertion (the present `frmp.tex` Assumption A3) is never used here.

2. The defining contour integral (Lewis/Parseval), proved

Proposition 2.1. Under (H), with $c \in (1, p^*)$,

$$C_M = \frac{K e^{-rT}}{2\pi i} \int_{c-i\infty}^{c+i\infty} \frac{\phi_M(-iw) e^{-w\tilde{k}}}{w(w-1)} dw. \quad (2.1)$$

Proof. The bilateral Laplace transform of the payoff $g_C(x) = S_0 e^{rT} (e^x - e^{\tilde{k}})^+$ is, for $\Re w > 1$,

$$\hat{g}_C(w) = \int_{\mathbb{R}} g_C(x) e^{-wx} dx = S_0 e^{rT} \int_{\tilde{k}}^{\infty} (e^x - e^{\tilde{k}}) e^{-wx} dx = S_0 e^{rT} \frac{e^{(1-w)\tilde{k}}}{w(w-1)},$$

both elementary integrals converging because $\Re(1-w) < 0$ and $\Re w > 0$. Parseval for the bilateral Laplace transform pairs g_C against the density p of X_T : for a vertical line in the common strip of analyticity of $\hat{g}_C$ (i.e. $\Re w > 1$) and of the moment transform $\mathbb{E}[e^{wX_T}] = \phi_M(-iw)$ (i.e. $\Re w < p^*$),

$$\mathbb{E}[g_C(X_T)] = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \hat{g}_C(w) \phi_M(-iw) dw.$$

The exchange is justified by absolute integrability on the line: $\hat{g}_C(w) = O(|w|^{-2})$ and, by §3.1 below, $|\phi_M(-iw)|$ has Gaussian decay. Insert $\hat{g}_C$, use $S_0 e^{rT} e^{\tilde{k}} = K$, and multiply by e^{-rT} . ■

The put follows from the *same integrand* on a line $c_P \in (-p^*, 0)$; $C_M - \Pi_M$ is the residue sum at the two fixed kernel poles $w \in \{0, 1\}$, which (using only $\phi_M(0) = \phi_M(-i) = 1$) equals $S_0 - K e^{-rT}$: put-call parity, independent of any pole structure. Hence it suffices to construct C_M .

3. One Gaussian times one entire exponential (Euclidean division — no poles)

3.1 The split is unconditional

Φ_M is a ratio of two explicit polynomials. Polynomial division — valid for **every** rational function, with no hypothesis whatsoever on the roots of the denominator — yields

$$\Phi_M(u) = -\frac{1}{2}\sigma_T^2 u^2 - i\mu_T u + \rho(u), \quad \rho = \frac{P}{D}, \quad \deg P < \deg D, \quad \rho(\infty) = 0, \quad (3.1)$$

with $\sigma_T^2 > 0$ a property *read off the two computed leading coefficients* (checkable; refuse the input if it fails — never silently patch). Set $\xi := \tilde{k} + \mu_T$ and

$$g(w) := \frac{1}{2}\sigma_T^2 w^2 - w\xi, \quad z_w := \frac{\xi - \sigma_T^2 w}{\sigma_T \sqrt{2}}.$$

Substituting $u = -iw$ into (3.1) (so $-\frac{1}{2}\sigma_T^2(-iw)^2 = \frac{1}{2}\sigma_T^2 w^2$, $-i\mu_T(-iw) = -\mu_T w$) and multiplying by $e^{-w\tilde{k}}$ gives the factorisation that drives everything:

$$\boxed{\phi_M(-iw) e^{-w\tilde{k}} = e^{g(w)} \psi(w), \quad \psi(w) := \exp(r(w)), \quad r(w) := \rho(-iw).} \quad (3.2)$$

On $\Re w = c$, $|e^{g(w)}| = e^{\sigma_T^2(c^2 - t^2)/2 - c\xi}$ decays like $e^{-\sigma_T^2 t^2/2}$ — the Gaussian decay used in §2. The factor $\psi = e^r$ is the **entire exponential** of a proper rational function: by (H) it is analytic and bounded on the closed half-plane $\{\Re w \leq c\}$ with $\psi(\infty) = 1$. Its singularities (images of whatever poles D has) all lie strictly in $\{\Re w > c\}$; **we never compute them**.

3.2 Why a naive single series cannot work — and what must replace it

The object to expand is ψ on the **infinite straight contour** $\Re w = c$. Any power series has a *disk* of convergence, and an infinite line never lies in a disk. Concretely:

- *Expand at $w = \infty$* ($\psi = \sum \hat{q}_n w^{-n}$): converges only for $|w| > \max_j |\hat{u}_j|$; the contour dips inside that disk near the real axis → **diverges on the contour**.
- *Expand at a finite point* (Gram-Charlier): radius = distance to the nearest pole; the contour runs to $\pm i\infty$ → **diverges**.

The present multi-index series dodges this by expanding each $\exp(B_j/(w - \hat{u}_j))$ separately (each is *entire* in $1/(w - \hat{u}_j)$, so converges off its own pole) and Cauchy-multiplying the $2M$ factors — which is exactly what produces the **non-enumerable lattice sum** $\sum_{\mathbf{n} \in \mathbb{Z}_{\geq 0}^{2M}}$. That is the disease.

The cure is to compactify the contour. Map the half-plane $\{\Re w < c\}$ conformally onto the unit disk; the infinite line becomes the unit **circle** (compact), and ψ , analytic in a slightly larger half-plane, becomes analytic in a disk of radius > 1 . *Then* one ordinary Taylor series converges uniformly on the whole contour. This is the “Taylor series of an entire exponential” realised honestly.

4. The Schwinger-Gauss-erfc kernel (proved from scratch, numerically certified)

Everything reduces to one contour integral, which we prove without reference to any prior text.

Lemma 4.1 (Schwinger-Gauss-erfc). *For $\sigma_T^2 > 0$ and $\Re w_* < c$,*

$$\mathcal{E}(w_*) := \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \frac{e^{g(w)}}{w - w_*} dw = \frac{1}{2} e^{g(w_*)} \operatorname{erfc}(z_{w_*}). \quad (4.1)$$

Proof. Since $\Re(w - w_*) = c - \Re w_* > 0$ on the line, write $\frac{1}{w - w_*} = \int_0^\infty e^{-(w - w_*)s} ds$ and exchange (Tonelli; the double integral of absolute values is finite by the Gaussian factor):

$$\mathcal{E}(w_*) = \int_0^\infty e^{w_* s} J(s) ds, \quad J(s) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} e^{g(w) - ws} dw.$$

Completing the square, $g(w) - ws = \frac{\sigma_T^2}{2} (w - \frac{\xi + s}{\sigma_T^2})^2 - \frac{(\xi + s)^2}{2\sigma_T^2}$. On $w = c + it$ with $b = c - \frac{\xi + s}{\sigma_T^2} \in \mathbb{R}$ and $\int_{\mathbb{R}} e^{-At^2 + iBt} dt = \sqrt{\pi/A} e^{-B^2/4A}$ ($A = \sigma_T^2/2$, $B = \sigma_T^2 b$),

$$J(s) = e^{-\frac{(\xi + s)^2}{2\sigma_T^2}} \cdot \frac{1}{2\pi} \int_{\mathbb{R}} e^{\frac{\sigma_T^2}{2} (b + it)^2} dt = \frac{1}{\sigma_T \sqrt{2\pi}} e^{-\frac{(\xi + s)^2}{2\sigma_T^2}}.$$

Now complete the square in s : $w_* s - \frac{(\xi + s)^2}{2\sigma_T^2} = -\frac{1}{2\sigma_T^2} (s - (\sigma_T^2 w_* - \xi))^2 + g(w_*)$, so with $u = (s - (\sigma_T^2 w_* - \xi)) / (\sigma_T \sqrt{2})$ (whence $s = 0 \mapsto u = z_{w_*}$),

$$\mathcal{E}(w_*) = \frac{e^{g(w_*)}}{\sigma_T \sqrt{2\pi}} \int_0^\infty e^{-\frac{(s - (\sigma_T^2 w_* - \xi))^2}{2\sigma_T^2}} ds = \frac{e^{g(w_*)}}{\sqrt{\pi}} \int_{z_{w_*}}^\infty e^{-u^2} du = \frac{1}{2} e^{g(w_*)} \operatorname{erfc}(z_{w_*}). \blacksquare$$

(For $\Re w_* > c$ shift the line past w_* ; the swept simple pole has residue $e^{g(w_*)}$, giving the extra $-e^{g(w_*)}$ recorded by the indicator in (4.3).)

Numerical certificate (independent of any source). Direct high-resolution contour integration of (4.1) vs the closed form, for $w_* \in \{0, 1, -0.4 + 0.6i, 0.2 - 0.9i\}$ at $c = 1.3$, agreed to $|\text{diff}| \leq 2 \times 10^{-16}$.

Lemma 4.2 (derivative atoms $\rightarrow$ Hermite functions). *Differentiating (4.1) $m - 1$ times in w_* (legitimate; the Gaussian dominates every derivative uniformly on the line),*

$$\mathcal{A}_m(w_*) := \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \frac{e^{g(w)}}{(w - w_*)^m} dw = \frac{1}{(m - 1)!} \partial_{w_*}^{m-1} \mathcal{E}(w_*). \quad (4.2)$$

With $z_w = (\xi - \sigma_T^2 w) / (\sigma_T \sqrt{2})$, the physicists' Hermite identity $\frac{d^j}{dz^j} \operatorname{erfc}(z) = -\frac{2}{\sqrt{\pi}} (-1)^{j-1} H_{j-1}(z) e^{-z^2}$ (immediate from $\operatorname{erfc}' = -\frac{2}{\sqrt{\pi}} e^{-z^2}$ and $\partial_z^n e^{-z^2} = (-1)^n H_n(z) e^{-z^2}$), together with the parameter-free cancellation $g(w) - z_w^2 = -\xi^2 / (2\sigma_T^2)$, expresses every $\mathcal{A}_m$ as $\operatorname{erfc}(z_{w_*})$

times a polynomial plus a finite sum of Hermite functions $H_{j-1}(z_{w_*})e^{-z_{w_*}^2}$ — closed form, no quadrature.

Numerical certificate. $\mathcal{A}_m(w_*)$ by contour integration vs (4.2) by Cauchy differentiation, for $w_* \in \{0, 1, 2.7, 2.7 + 0.5i\}$ (including $\Re w_* > c$, indicator on) and $m \in \{1, 2, 3, 4\}$: $|\text{diff}| \leq 7 \times 10^{-16}$ throughout.

The general node form, with $\mathbf{1}_{\{\Re w_* > c\}}$ for $\Re w_* > c$:

$$\mathcal{E}(w_*) = \frac{1}{2}e^{g(w_*)}\text{erfc}(z_{w_*}) - \mathbf{1}_{\{\Re w_* > c\}}e^{g(w_*)}. \quad (4.3)$$

The Hermite functions are generated by the two-term recurrence $P_0 \equiv 1$, $P_{k+1} = P'_k + (\sigma_T^2 w - \xi)P_k$ (so $\partial_w^k e^g = e^g P_k$).

5. The Cayley compactification

Fix a scale $\kappa > 0$ and define the Möbius map

$$\zeta(w) = \frac{w - c + \kappa}{w - c - \kappa} = 1 + \frac{2\kappa}{w - q}, \quad q := c + \kappa, \quad (5.1)$$

$$\boxed{w(\zeta) = c - \kappa \frac{1 + \zeta}{1 - \zeta}} \quad (5.2)$$

(The sign in (5.2) is not cosmetic: $w(0) = c - \kappa$ must lie **left** of the contour. Putting it right of the contour sends the disk onto the pole side and the series diverges — a mistake caught only by numerical validation, never by inspection.) The map sends

region in w	image in ζ
left half-plane $\{\Re w < c\}$ (where ψ is analytic)	open unit disk $\{ \zeta < 1\}$
contour $\{\Re w = c\}$	unit circle $\{ \zeta = 1\}$, minus $\zeta = 1 \leftrightarrow w = \infty$
right half-plane $\{\Re w > c\}$ (all surrogate poles)	exterior $\{ \zeta > 1\}$

Lemma 5.1 ($R_0 > 1$). *Every pole $\hat{u}$ of ψ (with $\Re \hat{u} \geq p^* > c$) satisfies $|\zeta(\hat{u})| > 1$, and $R_0 := \inf_{\hat{u}} |\zeta(\hat{u})| > 1$.*

Proof. For $a := \hat{u} - c$ with $\Re a > 0$, $|a + \kappa|^2 - |a - \kappa|^2 = 4\kappa \Re a > 0$, so $|\zeta(\hat{u})| = |a + \kappa|/|a - \kappa| > 1$. The infimum over the finitely many poles is attained and exceeds 1. ■

Hence $\Psi(\zeta) := \psi(w(\zeta))$ is analytic on $\{|\zeta| < R_0\}$ with $R_0 > 1$. **We never need the value of R_0 , only $R_0 > 1$ from (H).**

6. The single series

6.1 One Maclaurin series, uniformly convergent on the contour

By Lemma 5.1, Ψ has a Maclaurin series with geometric Cauchy bound

$$\Psi(\zeta) = \sum_{n=0}^{\infty} q_n \zeta^n, \quad |q_n| \leq \|\Psi\|_R R^{-n} \quad (\text{any } 1 < R < R_0), \quad (6.1)$$

converging **uniformly on** $|\zeta| = 1$ = the contour. One index n . No lattice. The dimension $2M$ has disappeared from the sum.

6.2 The coefficients by two single-index recurrences (no roots, no multi-index)

$r(w) = \rho(-iw)$ is rational, hence rational in ζ after substituting (5.2) and clearing $(1 - \zeta)^{\deg}$: $R(\zeta) := r(w(\zeta)) = A(\zeta)/B(\zeta)$. Because a rational function is **C-finite**, the identity $BR = A$ gives its Taylor coefficients $R(\zeta) = \sum_{m \geq 0} r_m \zeta^m$ by the linear recurrence $\sum_k b_k r_{m-k} = a_m$ — driven by the *coefficients* of B (the rotated Padé denominator), **never its roots**. Then $\Psi = e^R$ has coefficients by the classical exp-of-a-series recurrence ($\Psi' = R' \Psi$):

$$\boxed{q_0 = e^{r_0}, \quad n q_n = \sum_{m=1}^n m r_m q_{n-m} \quad (n \geq 1).} \quad (6.2)$$

Both are single-index ball-arithmetic convolutions — the same shape as the Chebyshev/Müntz recurrences already in arb4j.

6.3 The term integrals — erfc and Hermite at three fixed nodes

Split $\frac{1}{w(w-1)} = \frac{1}{w-1} - \frac{1}{w}$ and set, for $\delta \in \{0, 1\}$,

$$T_n^{(\delta)} := \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \frac{e^{g(w)} \zeta(w)^n}{w - \delta} dw, \quad T_0^{(\delta)} = \mathcal{E}(\delta). \quad (6.3)$$

From (5.1), $\zeta(w)^n = \sum_{k=0}^n \binom{n}{k} (2\kappa)^k (w - q)^{-k}$, so each $T_n^{(\delta)}$ is a finite combination of the atoms (4.2) at the **three fixed nodes** $\{0, 1, q\}$ — the payoff poles and the single Cayley pole, all chosen by us, none from the surrogate. Equivalently, the node recurrence (Appendix B.2) advances them without re-expanding:

$$T_n^{(\delta)} = T_{n-1}^{(\delta)} + \frac{2\kappa}{q - \delta} \left(T_{n-1}^{(q)} - T_{n-1}^{(\delta)} \right), \quad (6.4)$$

coupling $T^{(0)}, T^{(1)}, T^{(q)}$, each seeded from the fixed erfc/Hermite atoms (4.2)–(4.3) (the node $q = c + \kappa > c$ carries the indicator term).

6.4 The formula

By the uniform convergence of (6.1) against the absolutely integrable Gaussian weight $e^g/(w(w-1))$ (Theorem 7.1 licenses the exchange),

$$C_M = Ke^{-rT} \sum_{n=0}^{\infty} q_n (T_n^{(1)} - T_n^{(0)}), \quad \Pi_M = C_M - (S_0 - Ke^{-rT}). \quad (6.5)$$

One index. Coefficients q_n by (6.2); kernel terms by (6.3)–(6.4). The only special function is erfc . The surrogate enters **only** through $\{q_n\}$; the nodes z_0, z_1, z_q are fixed by σ_T, ξ, c, κ alone.

7. Convergence, error bound, and the sound stopping rule

7.1 Absolute convergence with a one-index geometric majorant

Theorem 7.1. *Under (H) and $\sigma_T^2 > 0$, the series (6.5) converges absolutely. With $G_\delta := \frac{1}{2\pi} \int_{\mathbb{R}} \frac{|e^{g(c+it)}|}{|c-\delta+it|} dt < \infty$ and any $1 < R < R_0$,*

$$\sum_{n \geq 0} |q_n| (|T_n^{(1)}| + |T_n^{(0)}|) \leq \|\Psi\|_R (G_0 + G_1) \sum_{n \geq 0} R^{-n} = \frac{\|\Psi\|_R (G_0 + G_1)}{1 - R^{-1}} < \infty. \quad (7.1)$$

Proof. On the contour $|\zeta(w)| = 1$, so straight from (6.3), $|T_n^{(\delta)}| \leq \frac{1}{2\pi} \int_{\mathbb{R}} \frac{|e^{g(c+it)}| |\zeta|^n}{|c-\delta+it|} dt = G_\delta$, **uniform in** n , finite because $|e^{g(c+it)}|$ decays like $e^{-\sigma_T^2 t^2/2}$. Combine with the Cauchy bound (6.1). Absolute convergence of the integrand sum, dominated by an L^1 Gaussian times $\sum R^{-n}$, licenses the term-by-term integration giving (6.5). ■

The bound uses **only** the Gaussian factor and $R_0 > 1$ (i.e. $p^* > c$): no pole is enumerated, located, or counted.

7.2 The a-posteriori tail bound is the stopping rule

Let $S_N = Ke^{-rT} \sum_{n=0}^N q_n (T_n^{(1)} - T_n^{(0)})$. By (7.1),

$$|C_M - S_N| \leq Ke^{-rT} (G_0 + G_1) \sum_{n > N} |q_n| \leq Ke^{-rT} (G_0 + G_1) \|\Psi\|_R \frac{R^{-(N+1)}}{1 - R^{-1}}. \quad (7.2)$$

Every quantity is a computable **ball**: $G_\delta \leq e^{\sigma_T^2 c^2/2 - c\xi}/(\sigma_T \sqrt{2\pi} |c - \delta|)$ (elementary), $\|\Psi\|_R \leq e^{\max_{|\zeta|=R} |R(\zeta)|}$ (one max of a rational function on a circle), and $R^{-(N+1)} \downarrow 0$ geometrically.

Stopping rule. Sum $n = 0, 1, 2, \dots$; after each term evaluate the right-hand ball of (7.2); **stop at the first N with that ball below the target radius 2^{-p}** ; return S_N widened by it. The result is a *certified enclosure* of C_M .

This is exactly the “compare the tail to the radius of the running ball” idea the commits 697f680d/fafbf3f reached for — but **sound**, because the tail is dominated by a *proven a-priori geometric majorant* (7.2), not by the empirical smallness of the last term. No best/min-term fallback, no Aitken, no $\sim$ sigil. The majorant is part of the algorithm’s specification, in full compliance with “prove it on its domain, deliver a verified ball.”

7.3 End-to-end numerical certificate (the decisive test against truth)

A concrete rational surrogate was built with $\sigma_T^2 = 0.6$, $\mu_T = \frac{1}{2}\sigma_T^2$, $S_0 = 100$, $K = 110$, $rT = 0.03$, and an arbitrary proper part $r(w) = \sum B_j/(w - \hat{u}_j)$ with four poles at $\Re \hat{u}_j \in \{2.5, 4.1\}$ (so $p^* = 2.5$), contour $c = 1.7 \in (1, p^*)$, $\kappa = 1$. The **truth** was the defining contour integral (2.1) evaluated by high-resolution quadrature; the **series** was (6.5) with q_n from (6.1)–(6.2) and $T_n^{(\delta)}$ from (6.3):

N	partial sum S_N	$ S_N - C_{\text{truth}} $
0	22.5709661629	4.4×10^0
1	17.9192435886	2.3×10^{-1}
2	18.1842148976	3.8×10^{-2}
5	18.1467027836	1.0×10^{-5}
10	18.1466952685	2.9×10^{-6}
20	18.1466923922	1.8×10^{-10}
40	18.1466923924	1.1×10^{-14}

with $C_{\text{truth}} = 18.1466923924$ and coefficients decaying geometrically ($|q_n|$: 1.7×10^{-1} , 6.0×10^{-3} , 3.1×10^{-4} , 1, ratio $\rightarrow R^{-1} \approx 0.5$). This is the mathematical ground truth: the single series **is** the price, and it converges geometrically. (Beyond $N \approx 45$ a *crude* FFT coefficient extraction hits roundoff; the exact recurrences (6.2) in ball arithmetic do not — that is an artifact of the validation script, not the construction.)

7.4 Convergence in the Padé order M

The fixed- M result (U) above is self-contained. Whether $C_M \rightarrow C$ as $M \rightarrow \infty$ is the conditional (C) statement (Stahl); it is orthogonal to the single-series question and untouched here.

8. The Black-Scholes certificate (the $n = 0$ term)

When the proper part vanishes ($r \equiv 0$): $R \equiv 0$, so $q_0 = e^0 = 1$ and $q_n = 0$ for $n \geq 1$. The series (6.5) collapses to

$$C_M = Ke^{-rT}(T_0^{(1)} - T_0^{(0)}) = Ke^{-rT}(\mathcal{E}(1) - \mathcal{E}(0)).$$

With $\mu_T = \frac{1}{2}\sigma_T^2$ (martingale): $\mathcal{E}(0) = \frac{1}{2}\text{erfc}(z_0) = N(-\xi/\sigma_T) = N(d_2)$ and $Ke^{-rT}\mathcal{E}(1) = Ke^{-rT}\frac{S_0}{K}e^{rT}N(d_1) = S_0N(d_1)$, using $N(x) = \frac{1}{2}\text{erfc}(-x/\sqrt{2})$, $g(1) = -\tilde{k}$, $e^{-\tilde{k}} = \frac{S_0}{K}e^{rT}$, and $-\xi/\sigma_T = d_2$, $-(\xi - \sigma_T^2)/\sigma_T = d_1$. Hence

$$C_M = S_0 N(d_1) - K e^{-rT} N(d_2), \quad d_{1,2} = \frac{\log(S_0/K) + rT}{\sigma_T} \pm \frac{\sigma_T}{2}.$$

Exactly Black-Scholes — now literally the **first term** of the one series, not a degeneration of a lattice. This fixes $\xi = \tilde{k} + \mu_T$ and the prefactor $K e^{-rT}$ with no free constant.

9. What the post-last_known_good attempt did, and why it is wrong

Measured against the proven construction above (not against any source text):

1. **Multi-index price (frmp.tex Thm price, eq Cprice).** A sum over $\mathbb{Z}_{\geq 0}^{2M}$, produced by Cauchy-multiplying $2M$ separate pole-exponential series. It is **not enumerable as one series** and so cannot be summed by $\Sigma\{n=0.. \infty\}$. The single series (6.5) replaces it; the $2M$ survives only as the fixed *length* of the recurrence (6.2).
 2. **Assumed pole structure (frmp.tex A3 / Prop strip).** “ $2M$ simple poles, all in the lower half-plane, proved for large M ” is a structural, asymptotic, generic claim about the roots of a computed polynomial. The price must not depend on it. §2–§7 use only **Euclidean division** (always valid) and **(H)** (a contour exists — necessary for finiteness). A3 is discarded entirely.
 3. **Heuristic truncation (arb.functions.ConvergentSeriesAccumulator).** It stops when $|\text{term}| \leq 2^{\{-\text{bits}/2\}}$ and otherwise returns a min-term “best” partial sum. Neither is a verified tail enclosure: for any series whose terms shrink-then-grow it can stop early and return a ball that does **not** contain the truth, and the “best” return is a forbidden fallback. The geometric majorant (7.2) makes truncation sound and retires this class entirely — along with the $\sim$ sigil, Aitken- Δ^2 , optimallyTruncated, and the SCC generation-time dispatch (all accelerations/special-cases, none an enclosure).
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10. Implementation in arb4j — make the source conform to the proven construction

The single series (6.5) is **one** $\Sigma\{n=0.. \infty\}$ over three context-shared Sequences plus a verified tail. The correctness target is the mathematics of §4–§8, validated as in §7.3 — not the present source.

Build:

1. **Proper part r .** From the diagonal $[M/M]$ Padé of the cgf Müntz series (MuntzPadeApproximant), take numerator/denominator and run **one** GenericRingPolynomial.divrem to get σ_T^2, μ_T, ρ as in (3.1). Assert $\sigma_T^2 > 0$; **throw** otherwise (explicit refusal, never a silent patch).
2. **Cayley coefficients r_m** by the C-finite recurrence of §6.2 (substitute (5.2), clear denominators with ComplexPolynomial arithmetic). One Sequence.
3. **Exponential coefficients q_n** by (6.2). One Sequence of Complex balls — the same convolution the compiler emits for $\Sigma k \rightarrow x^k/k!$.
4. **Kernel atoms $\mathcal{E}(\delta), \partial^m \mathcal{E}(q)$** at the fixed nodes $0, 1, q$ from acb_hypgeom_erfc and the two-term Hermite recurrence $P_{k+1} = P'_k + (\sigma_T^2 w - \xi) P_k$. One Sequence per node.
5. **Term sequence $t_n = q_n(T_n^{(1)} - T_n^{(0)})$** via (6.4); sum with **one** $\Sigma\{n=0.. \infty\}$.

6. **Verified-tail accumulator.** A generic accumulator taking the term sequence **and the a-priori majorant** $\tau(N) = \text{RHS of (7.2)}$; stop when $\tau(N) \leq 2^{-p}$; return the ball $S_N \pm \tau(N)$.

Compiles to one expression of shape $n \rightarrow q(n) \cdot (T_1(n) - T_0(n))$ $\{n=0.. \infty\}$ with q, T_1, T_0 wired top-down through the context (the ordinary recursion machinery), summed by the verified $\Sigma\{.. \infty\}$.

Delete: the multi-index price assembly; Assumption A3 / Prop *strip* and the pole-list rotation/multinomial lemmas; ConvergentSeriesAccumulator's $2^{\{-\text{bits}/2\}}$ stop and best return; the $\sim$ sigil, Aitken- Δ^2 , optimallyTruncated, SCC dispatch. MuntzPadeCumulantGenerator need not return: the cgf is $\Sigma\{k=0.. \infty\} d_k(v) z^k$ and the price is the new $\Sigma\{n=0.. \infty\}$ of (6.5) — two ordinary infinite sums sharing one verified-tail primitive.

The one new primitive: a $\Sigma\{.. \infty\}$ whose convergence test consults an a-priori majorant $\tau(N) \downarrow 0$ attached to the summand, not the magnitude of the last term. Every absolutely convergent series `arb4j` cares about (this price, e^x , Mittag-Leffler, Fox-H) carries such a majorant; making it a first-class part of the summand contract is the single principled change that retires the whole heuristic tower.

Validate the implementation the way §7.3 validates the math: (i) exact Black-Scholes on $r \equiv 0$ to full precision; (ii) cross-check against the defining contour integral (2.1) by independent quadrature on random admissible surrogates; (iii) confirm geometric coefficient decay and that the certified ball (7.2) actually contains the quadrature value. Agreement with the present source is **not** a validation criterion.

Appendix A. Standing facts used (all elementary)

$$\operatorname{erfc}(z) = \frac{2}{\sqrt{\pi}} \int_z^\infty e^{-u^2} du; \quad \int_{\mathbb{R}} e^{-At^2 + iBt} dt = \sqrt{\pi/A} e^{-B^2/4A} \quad (\Re A > 0); \quad N(x) = \frac{1}{2} \operatorname{erfc}(-x/\sqrt{2}); \quad \partial_z^n e^{-z^2} = (-1)^n H_n(z) e^{-z^2} \text{ (physicists' } H_n \text{)}.$$

Appendix B. Deferred proofs

B.1 ($R_0 > 1$). §5, Lemma 5.1: $|a + \kappa|^2 - |a - \kappa|^2 = 4\kappa \Re a > 0$ for $a = \hat{u} - c$, $\Re a \geq p^* - c > 0$. ■

B.2 (node recurrence (6.4)). From $\zeta = 1 + \frac{2\kappa}{w-q}$, $\zeta^n = \zeta^{n-1} + \frac{2\kappa}{w-q} \zeta^{n-1}$. Divide by $w - \delta$, integrate against $\frac{1}{2\pi i} e^g$, and use $\frac{1}{(w-q)(w-\delta)} = \frac{1}{q-\delta} \left(\frac{1}{w-q} - \frac{1}{w-\delta} \right)$, giving $T_n^{(\delta)} = T_{n-1}^{(\delta)} + \frac{2\kappa}{q-\delta} (T_{n-1}^{(q)} - T_{n-1}^{(\delta)})$, with $T_n^{(q)}$ the same family at node q (whose atoms (4.3) carry the $\mathbf{1}_{\{\Re q > c\}}$ term, $q = c + \kappa > c$). ■

B.3 (exp recurrence (6.2)). $\Psi = e^R \Rightarrow \Psi' = R' \Psi$; matching ζ^{n-1} coefficients in $\sum n q_n \zeta^{n-1} = (\sum m r_m \zeta^{m-1}) (\sum q_k \zeta^k)$ gives $n q_n = \sum_{m=1}^n m r_m q_{n-m}$. ■

Appendix C. Reproduction of the §7.3 certificate

Truth = quadrature of (2.1); series = (6.5) with (6.1)–(6.3). Independent quadrature of (4.1) and (4.2) confirmed Lemmas 4.1–4.2 to $\leq 7 \times 10^{-16}$. Key correctness guards a reimplementations must reproduce: the inverse-map sign (5.2); nodes $\{0, 1, q\}$ are fixed (never the surrogate poles); $|q_n|$ decays at rate $R^{-1} < 1$; the certified ball (7.2) encloses the quadrature value.